Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026

Mid-Semester Test
EC-2 Makeup

Course No. : AIMLCZC416/DESC ZC418
Course Title : MFML/NFDS
Nature of Exam : Closed Book
Weightage : 30%

Q1.(a) Find the Echelon form of the matrix $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$ and hence find the determinant of A. [2M]

Solution:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \approx \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \approx \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$ This is Echelon form of A[1M]

Determinant of A = 1 * 1 * 1 * 1 = 1 [1M]

(b) Find the inverse of the matrix A, and using this inverse solve the system of linear equations

$AX = b \text{ where } b = \begin{bmatrix} -2 \\ 2 \\ 1 \\ -1 \end{bmatrix}$. [2M]

Solution:

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right) \approx \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\approx \left(\begin{array}{cccc|cccc} 1 & 0 & -1 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right)$$ [0.5M]

$$\approx \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 1 & 0 & -1 & -2 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 4 & -3 & 2 & -1 \\ 0 & 1 & 0 & 0 & -3 & 3 & -2 & 1 \\ 0 & 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array}\right)$$

Hence the inverse of A = $\begin{pmatrix} 4 & -3 & 2 & -1 \\ -3 & 3 & -2 & 1 \\ 2 & -2 & 2 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}$ [1M]

The solution of the system AX = b is

$$X = \begin{pmatrix} 4 & -3 & 2 & -1 \\ -3 & 3 & -2 & 1 \\ 2 & -2 & 2 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix} \begin{bmatrix} -2 \\ 2 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} -11 \\ 9 \\ -5 \\ 2 \end{bmatrix} \quad [0.5M]$$

Q2. Consider the matrix $A = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 2 & -2 \\ 3 & -3 & 3 \end{bmatrix}$.

(a) Show that the set $V = \{X \in R^3 : AX = 0\}$ is a subspace of R^3. [2M]

Solution:

To show that a set V is a subspace, we must verify whether it is closed with respect to vector addition and scalar multiplication.

If $X_1, X_2 \in V$, then **(1M)**

$$AX_1 = 0, AX_2 = 0$$
$$A(X_1 + X_2) = AX_1 + AX_2 = 0$$
$$\text{So, } X_1 + X_2 \in V.$$

For any scalar c and $X \in V$,

$$A(cX) = c(AX) = c \cdot 0 = 0$$
$$\text{Hence, } cX \in V. \qquad (1M)$$

Therefore, V is a subspace of R^3

[If they only check V is closed under vector addition and scalar multiplication please give full 2 marks]

(b) Determine a basis for V and hence find its dimension. [2M]

Solution:

We solve the homogeneous system $AX = 0$.

Let $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, From $AX = 0$

$$x - y + z = 0$$
$$-2x + 2y - 2z =$$
$$3x - 3y + 3z = 0$$

All three equations are **scalar multiples** of the first equation. So, we only need:

$$\boldsymbol{x - y + z = 0}$$

Solving for x:

$$x = y - z$$

Writing the solution in parametric form:

$$X = \begin{bmatrix} y - z \\ y \\ z \end{bmatrix} = y\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$ **(1M)**

Basis of V: $\begin{bmatrix} \mathbf{1} \\ \mathbf{1} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{-1} \\ \mathbf{0} \\ \mathbf{1} \end{bmatrix}$**, Dimension of V:2** **(0.5M +0.5M)**

(c) Show that V has a one-dimensional subspace U and find a basis for U. [2M]
Solution:

Since dim $(V) = 2$, any **non-zero vector** in V spans a **one-dimensional subspace**.

Choosing one vector from the basis of V, u= $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Since this is a single non-zero vector it is linearly independent.

Also we know from a standard theorem that U = span $\left\{\begin{bmatrix} \mathbf{1} \\ \mathbf{1} \\ \mathbf{0} \end{bmatrix}\right\}$ ***is a subspace of V.[0.5M]***

Define: $\boldsymbol{U} = \textbf{span}\left\{\begin{bmatrix} \mathbf{1} \\ \mathbf{1} \\ \mathbf{0} \end{bmatrix}\right\}$ **(0.5M)**

Then: $U \subset V$, dim $(U) = 1$

Basis of U: $\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right\}$ **(1M)**

Note: considering another nonzero vector is also correct

Q3. Find the singular value decomposition of the matrix $A = \begin{bmatrix} 5 & 0 \\ 5 & 0 \\ 0 & -10 \end{bmatrix}$. [6M]

Q3 Find the Singular Value decomposition of the matrix $A = \begin{bmatrix} 5 & 0 \\ 5 & 0 \\ 0 & -10 \end{bmatrix}$ [6M]

Sol:- $A^TA = \begin{bmatrix} 50 & 0 \\ 0 & 100 \end{bmatrix}$ $\sigma_1 = \sqrt{100} = 10, \; \sigma_2 = \sqrt{50}$ (1M)

For $\lambda = 100 \Rightarrow \begin{bmatrix} 50-100 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (0.5M)

$\lambda = 50 \Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 50 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (0.5M)

$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (0.5M)

$u_1 = \frac{1}{10} A v_1 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{50}} \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$ (0.5+0.5)

Let u_3 be orthogonal to u_1 and u_2 (0.5)

Let $u_3 = (x, y, z) \Rightarrow \langle u_3 \; u_1 \rangle = 0 \Rightarrow z = 0,$

$\langle u_3 \; u_2 \rangle = 0 \quad \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = 0 \quad \therefore u_3 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$ (0.5)

$U = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1 & 0 & 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 10 & 0 \\ 0 & 5\sqrt{2} \\ 0 & 0 \end{bmatrix}$ 0.5+0.5

$\therefore \boxed{A = U \Sigma V^T}$ (0.5)

Q4. Let $A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$.

(a) Prove that A is a symmetric, positive definite matrix. [2M]

(b) Using this matrix A, define an inner product on R^4. Using this inner product, compute the distance between $\begin{bmatrix} -2 \\ 0 \\ 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 2 \\ 1 \\ 0 \end{bmatrix}$. [2M]

(c) a) **To prove that A is symmetric**

Symmetry

Since $A^T = A$, the matrix is **symmetric**.

(d) **(0.5M)**

(e) To prove that A is a positive definite matrix (Any one method….(1.5M)

(f) Method 1:

Positive definiteness

Because A is symmetric, it is positive definite iff all **leading principal minors** are positive.

1. First minor:

$$\det[2] = 2 > 0$$

2. Second minor:

$$\det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 6 - 1 = 5 > 0$$

(g)

3. Third minor:

$$\det \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} = 4 \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 4 \times 5 = 20 > 0$$

(h)

4. Fourth (full determinant):

   The matrix is block diagonal with blocks $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ and $\begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$.

$$\det A = 5\,(8 - 1) = 35 > 0$$

Hence, **all leading principal minors are positive**, so

$$\boxed{A \text{ is positive definite.}}$$

(i)

(j) Method 2: Quadratic form

For a symmetric matrix, **positive definiteness** can be proved by showing

$$x^T A x > 0 \quad \text{for all } x \neq 0.$$

Let

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}.$$

(k)

Step 1: Compute the quadratic form

$$x^T A x = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \end{pmatrix} A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

$$= 2x_1^2 + 2x_1x_2 + 3x_2^2 + 4x_3^2 + 2x_3x_4 + 2x_4^2$$

(l)

$$= (x_1^2 + 2x_1x_2 + x_2^2) + x_1^2 + 2x_2^2 + (x_3^2 + 2x_3x_4 + x_4^2) + 3x_3^2 + x_4^2$$

$$= (x_1 + x_2)^2 + x_1^2 + 2x_2^2 + (x_3 + x_4)^2 + 3x_3^2 + x_4^2$$

(m)

Each term is **non-negative** and equals zero **only when**

$$x_1 = x_2 = x_3 = x_4 = 0.$$

Hence

$$\boxed{x^T A x > 0 \quad \forall x \neq 0}$$

$$\Rightarrow \boxed{A \text{ is positive definite}}$$

(n)

Define the inner product using A:

$$\langle x, y \rangle_A = x^T A y$$

Definition (0.5M)

The induced distance is

$$d(x, y) = \sqrt{\langle x - y, x - y \rangle_A}.$$

Let

$$u = \begin{pmatrix} -2 \\ 0 \\ 1 \\ -1 \end{pmatrix}, \qquad v = \begin{pmatrix} -2 \\ 2 \\ 1 \\ 0 \end{pmatrix}.$$

$$d = u - v = \begin{pmatrix} 0 \\ -2 \\ 0 \\ -1 \end{pmatrix} \qquad Ad = \begin{pmatrix} -2 \\ -6 \\ -1 \\ -2 \end{pmatrix}$$

$$\langle d, d \rangle_A = d^T Ad = (0, -2, 0, -1) \cdot (-2, -6, -1, -2) = 14$$

$$\boxed{d(u, v) = \sqrt{14}}$$

Distance (1.5M)

Q5. Let $f(x, y, z) = xy^2 + yz^2 + zx^2$

(a) Find the gradient of $f(x, y, z)$. [2M]
(b) Find all the critical points of the function $g(x, y) = f(x, y, 0)$ [1M]
(c) Examine whether the second derivative (Hessian) test can be applied to classify the critical points of $g(x, y)$. Justify your answer. [2M]

Given:

$$f(x, y, z) = xy^2 + yz^2 + zx^2$$

(a) Gradient of f(x, y, z) [2 Marks]

$$\partial f/\partial x = y^2 + 2zx$$

$$\partial f/\partial y = 2xy + z^2$$

$$\partial f/\partial z = 2yz + x^2$$

$$\nabla f(x, y, z) = (y^2 + 2zx, 2xy + z^2, 2yz + x^2)$$

Partial derivatives – 1½ M
Gradient vector – ½ M

(b) Critical points of g(x, y) = f(x, y, 0) [1 Mark]

$$g(x, y) = xy^2$$

$$\partial g/\partial x = y^2$$

$$\partial g/\partial y = 2xy$$

Critical points: (x, 0), for all real x

g(x, y) – ½ M
Critical points – ½ M

(c) Hessian Test [2 Marks]
Second-order derivatives:

$$g_{xx} = 0, \; g_{yy} = 2x, \; g_{xy} = 2y$$

Hessian determinant:

$$D = g_{xx}g_{yy} - (g_{xy})^2$$

$$At \; (x, 0): D = 0$$

Conclusion: Hessian test is inconclusive.

Second derivatives – 1 M
Hessian & conclusion – 1 M

Q6. (a) Find the Taylor polynomial of degree 2 of the function $f(x) = \exp\,(\sin x^2)$ at 0. [4M]

(b) Just by observing the function f(x) defined in (a), write down the smallest and largest values of f(x). [1M]

The Taylor polynomial requires derivatives of $f(x) = \exp(\sin x^2)$ at $x = 0$.

Compute $f(0) = \exp(\sin 0) = 1$.
Find $f'(x) = \exp(\sin x^2) \cdot \cos(x^2) \cdot 2x$, so $f'(0) = 0$.
$f''(x)$ involves product and chain rules; evaluate at 0:
Only the term from derivative of $2x\cos(x^2) \cdot e^{\sin(x^2)}$ at $x = 0$ matters:
$f''(0) = 2\cos(0) \cdot e^0 = 2$.

Thus $P_2(x) = 1 + 0 \cdot x + \frac{2}{2!}x^2 = 1 + x^2$.

Since $-1 \leq \sin(x^2) \leq 1$, $\exp(\sin x^2)$ ranges from e^{-1} to e.
Smallest value e^{-1}, largest value e.

Computing derivatives -2 mark

Finding value of the function and derivatives at x=0: 1 mark

Writing Taylors polynomial: 1 mark

Writing smallest and largest value: 1 mark